

Loan Approval Prediction

Nicholas Mipro

Loan Approval

- Some of these aspects should not matter at all
 - Gender, age, marital status
 - Decided to drop occupation

```
1 df.head()
```

```
✓ 0.0s
```

	age	gender	occupation	education_level	marital_status	income	credit_score	loan_status
0	32	Male	Engineer	Bachelor's	Married	85000	720	Approved
1	45	Female	Teacher	Master's	Single	62000	680	Approved
2	28	Male	Student	High School	Single	25000	590	Denied
3	51	Female	Manager	Bachelor's	Married	105000	780	Approved
4	36	Male	Accountant	Bachelor's	Married	75000	710	Approved

The Data

- There looks to be no outliers

```
1 df.describe()
```

✓ 0.0s

	age	income	credit_score
count	61.000000	61.000000	61.000000
mean	37.081967	78983.606557	709.836066
std	8.424755	33772.025802	72.674888
min	24.000000	25000.000000	560.000000
25%	30.000000	52000.000000	650.000000
50%	36.000000	78000.000000	720.000000
75%	43.000000	98000.000000	770.000000
max	55.000000	180000.000000	830.000000

The Data

- There are no duplicates or missing values

```
1 df.duplicated().sum()
```

✓ 0.0s

```
np.int64(0)
```

```
1 df.isna().sum()
```

```
2 #No missing values
```

✓ 0.0s

age	0
gender	0
occupation	0
education_level	0
marital_status	0
income	0
credit_score	0
loan_status	0
dtype:	int64

Encoded and Scaled

	age	gender	education_level	marital_status	income	credit_score	loan_status
0	32	0	2	1	85000	720	0
1	45	1	3	0	62000	680	0
2	28	0	0	0	25000	590	1
3	51	1	2	1	105000	780	0
4	36	0	2	1	75000	710	0

	age	gender	education_level	marital_status	income	credit_score
0	0.258065	0.0	0.50	1.0	0.387097	0.592593
1	0.677419	1.0	0.75	0.0	0.238710	0.444444
2	0.129032	0.0	0.00	0.0	0.000000	0.111111
3	0.870968	1.0	0.50	1.0	0.516129	0.814815
4	0.387097	0.0	0.50	1.0	0.322581	0.555556

Linear Regression

- Linear Regression
 - Good fit with R^2 and RMSE

Linear Regression Mean Squared Error: 0.0596

Linear Regression R^2 Score: 0.7202

KNN

- KNN
 - High Accuracy, with good RMSE and ok R^2

KNN Accuracy: 0.9230769230769231

KNN Mean Squared Error: 0.0769

KNN R^2 Score: 0.6389

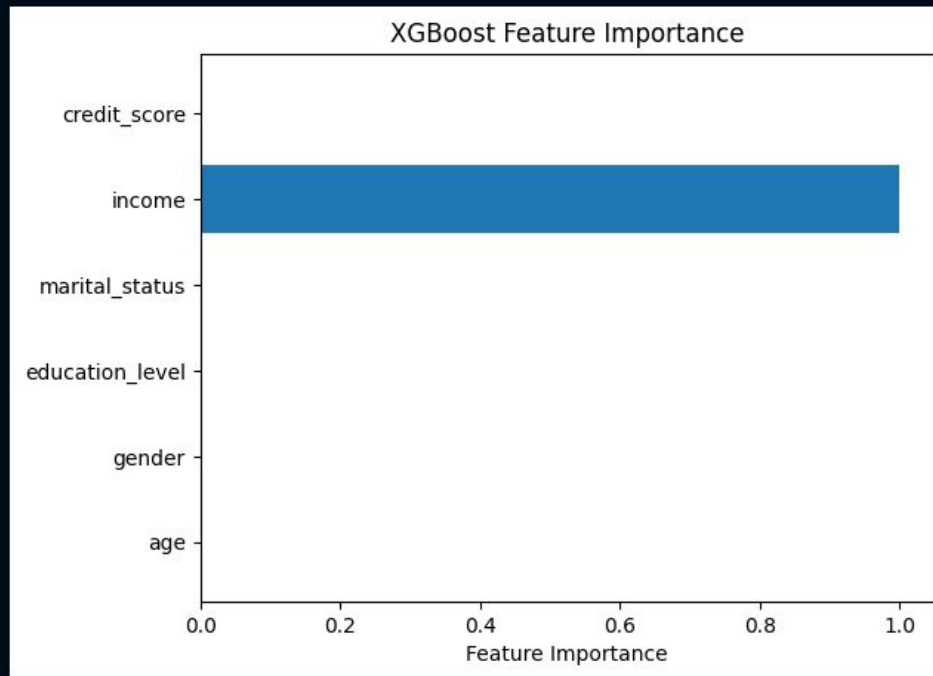
XGBoost

- Due to the limited amount of data I believe that XGBoost is memorizing the data

XGBoost Accuracy: 1.0000

XGBoost Mean Squared Error: 0.0000

XGBoost R² Score: 1.0000



Neural Network

- Neural Network
 - Bad R^2 score and good RMSE

```
loss = 0.055937591940164566
```

```
loss = 0.05524791032075882
```

```
loss = 0.054567739367485046
```

```
loss = 0.053898461163043976
```

```
loss = 0.05323963984847069
```

```
loss = 0.05259057879447937
```

```
Neural Network Accuracy: 0.8000
```

```
Neural Network Mean Squared Error: 0.2000
```

```
Neural Network  $R^2$  Score: -0.2500
```

Random Forest

- Random Forest has a good accuracy and demonstrates a good feature balance
- It actually give us some good insight

Random Forest Mean Squared Error: 0.0769

Random Forest R² Score: 0.6389

Random Forest Accuracy: 0.9231

